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Dissertation (C-uppsats)

Competitive Advantage of Environmental Sustainability

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Executive Summary

Title: Competitive Advantage of Environmental Sustainability

Unit: BA Bachelor Dissertation, 10p (15 ECTS)

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Background: More and more companies are trying to adopt a sustainability strategy, because of a growing awareness among people about a need for better environment in the future. It has been noticed that a balance between economic, social and environmental aspects is of great significance. The benefits of a strategy to become more balanced are said to be competitive advantage and stakeholder satisfaction besides the economic, social and environmental benefits.

Purpose: The purpose of the study is to find out the incentives for oil companies to implement a sustainable development strategy in order to be environmentally friendly, and whether it reaps competitive advantage.

Methodology: A qualitative approach has been used for gathering significant information for the study. Three oil companies have been investigated and five interviews has been made, to find out relevant data.

Theory: The theory delves deeper into the central subjects of discussion. The two concepts, sustainable development and competitive advantage has been explored in order to find the linkage between them.

Empirical Data: In this section the interviews are presented. The interviewed objects are the three companies Shell, Statoil and Preem and the five persons interviewed are the environmental director at Shell, the environmental co-ordinator and chief accountant at Preem as well as the senior vice president of environment and CSR (corporate social responsible) at Statoil.

Analysis: In this chapter the theoretical data has been compared to the empirical data. The sustainable development is subcategorised into environmental, economic and social aspects and focus on the competitive advantage of the strategy.

Conclusion: Finally, the conclusion will include the outcome of the study. It will show whether the purpose has been achieved or not.

TABLE OF CONTENTS

TABLE OF FIGURES

1. Introduction

The Authors will in the following chapter explain what the dissertation is about, how and why the research topic has been chosen, and an explanation of the basic concepts will be pointed out as well. The background includes a description of the problem that will lead to the main question and purpose of the work.

1.1 Background

If companies do not adapt Sustainable Development, which is considered as one of the most important environmental issues, then the companies will be financially affected through high fines that could lead to developmental problems. Furthermore, it will also affect the environment and the social atmosphere of the world, since the companies will "do bad things" (Schwartz, 1999). Companies are further pressured by Governments and Greenpeace, which affect them through legislation and bad publicity, to adapt this concept. Moreover, since the environment is vital and very important for future generations, there is a need for the economy of industries and life itself to apply the sustainable development (Elliot, 2006). Though, by being friendly to the environment, companies can grow and make profit by improving energy efficiency and lowering their production costs (Shell chemical Ltd, 2006).

SD → adds value to the company → environmentally friendly → CA → Economic growth

Figure 1 Adopting Sustainable Development Model (own illustration)

The idea of adopting the environment to sustainable development did not occur until the 1980s (Johnson and Turner, 2006) as a result of the awareness of the impacts of oil, gas and coal emissions on the atmosphere and sea. The gas and oil as two important commonly used kinds of fuel affect the atmosphere by producing carbon dioxide (CO_2), while polluted water runs into the seas and damages the sea life (Soares, 1999). As a consequence, high fines have been imposed by legislation in many countries. The European Union has ever since been the leader of the environmental policy, which partly involved the development of ambitious controls in areas such as climate change and the promotion of themes such as sustainable development in international fields (Jordan *et al.*, 2003)

The report of the World Commission on Environment and Development in 1987 (also known as the Brundtland Report) included the human development part, which brought attention to the challenges to overcome poverty and integration of environment in the decision-making of the economy. It further considered the political and social requirements of sustainable development in practise; hence if the environment would be taken care of, then it would gain economic growth (Elliot, 2006).

The EU adopted a completed sustainable development strategy (SDS) in 2006, which is built upon the 'Gothenburg strategy of 2001' and results from the 2004 start of an excessive review project (European Commission, 2007a). Big companies, such as Shell Oil Company, have adopted their own sustainable development strategies to satisfy stakeholders' needs and develop better competitive advantage. This implementation by Shell was made in 1997, as the result of the pressure practised by governments and Greenpeace. This, in fact, resulted from two main mistakes made by the company: the first was the decommissioning of the Brent Spar storage tank, which was environmentally hazardous, and, the second was the involvement in the trial and hanging of nine Ogoni tribe members in Nigeria. However, the company, along with others, still has a long way to go in order to achieve a complete SDS (Schwartz, 1999).

1.2 Thesis / Research question

How do oil companies gain competitive advantage of sustainable development and protecting the environment?

1.3 Purpose

The core purpose of this study is to gain a clear view of the competitive advantage with environmental sustainability. As a result of adapting a sustainable development strategy, companies will increase their long term profitability. In addition, companies will gain stakeholders satisfaction. With this study, the authors try to examine how the sustainability can positively effect the environment, economy and society (see figure 2). Moreover, it will be discussed whether sustainability lead to adding value to companies (see figure 1).

1.4 Concepts

Sustainable Development

As ICLEI (quoted in Mawhinney) states, sustainable development "delivers basic environmental, social and economic services to all residents of a community without threatening the viability of the natural, built and social systems upon which the delivery of these services depends" (2002, p.3).

Elliot presents another definition, which state, "sustainable development is a moral concept that seeks to define a 'fair and just' development" (2006, p.14).

Competitive Advantage

A significant competitive advantage is an advantage that organisations gain over their competitors, which provides great value to the customers and great benefit to the company (Lynch, 2006).

Competitive advantage can be gained either by lowering the prices or differentiating the products. It is possible as well to use both. While being sustainable, companies can maintain their competitive advantage in the market (Johnson and Scholes, 2002).

Environmental issues

The authors refer to environmental issues as local pollution, resource depletion, global pollution, climate change and loss of animal species. While the globalisation has accelerated the speed of change and the awareness of the environmental issues has risen (Mawhinney, 2002).

PESTEL analysis

A PESTEL analysis is used as a checklist by organisations to analyse the aspects that are imposed on them as political, economical, socio-cultural, technological, environmental and legal. This analysis relies on past events and experiences which are to be implemented in future strategic actions. It analyses the environmental pressure on the organisation (Lynch, 2006).

Stakeholder analysis

It is a technique, which can be used to identify and assess the importance of the people with interest in the organisation. The stakeholders are individuals and groups that can influence the organisations 'mission, objectives and strategies' (Lynch, 2006).

1.5 Strategies

The authors have selected two different strategies to clarify and describe the relation between the environmental, economic and social issues (see Figure 2) and how the interaction of these issues can maintain sustainable development.

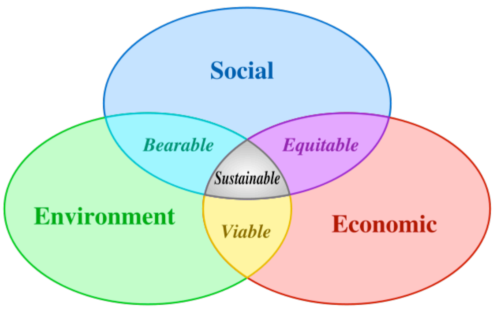
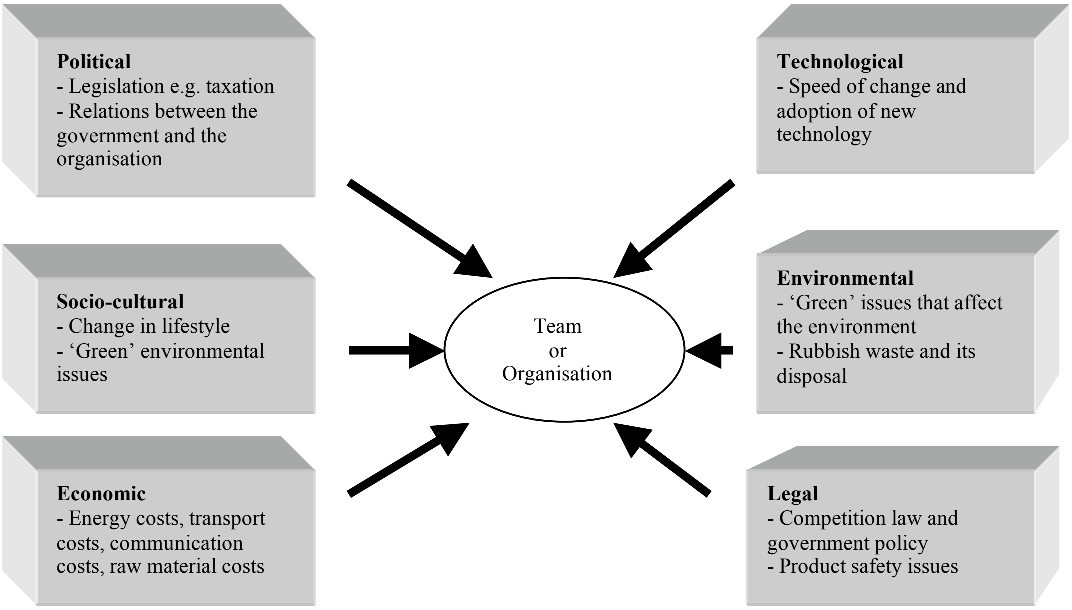


Figure 2, the objectives of SD, (Wikipedia, 2007)

These three pillars need to be addressed by government, businesses and civil society, since they inter-link considerable contribution to sustainable development (European Commission, 2001).

1.5.1 PESTEL analysis

The aim of this strategy is to show how the external factors affect the organisation and; therefore, the authors have chosen to use this analysis for their research (see Figure 3).



Figur 3 Factors affecting an organisation(Own illustration)

The European Commission has set up the Sixth Environment Action Programme as legislation for the EU member states. The governments in these member states can then further choose to set even stricter regulations on the companies to encourage them to "undertake environmental controls through the creation of economic incentives (via the tax systems) to reduce pollution" (Elliot, 2006, p.116) (see section 1.1 for further information about EU policy and pollution). This programme contributes to the environmental components for sustainable development, which places the policy of the environment in a broad perspective, and takes into consideration the social and economic aspects as well (see figure 2). This figure shows that there is a link between environment and European objectives for competitiveness and growth (European Commission, no date).
More government pressure on the companies by enforcements or penalties is anticipated, if they do not apply the imposed regulations. Therefore, the companies have to be environmental sustainable in order to avoid facing penalties (Elliot, 2006).

Companies should consider and understand the environmental changes, which have occurred lately and seize it as a key factor to create a competitive advantage by implementing a new sustainable strategy (Walsh, 2005). As noticed, people have become more aware of certain products that cause damage to the environment. These damages are caused during the process of manufacturing, use or disposal of products (Rock, 1989). As a result of this awareness companies have to be more environmental friendly and adapt to the new strategies that reduce the disposal impact on the environment and thereby could gain benefit from new economic and social processes (Elliot, 2006). The companies have also to take into consideration how to handle the disposal and waste, which affect the environment (see section 1.1) and; therefore; they have to investigate which positive outcomes the sustainable development would bring to both the company and the environment.

If companies give more attention to saving energy costs, they will not only benefit the environment, but can also bring financial advantages to the companies and they will improve their relationship with local communities, customers and suppliers (Carty, 1996).

Stricter environmental regulations do not lead to cost burdens for companies, but instead they could be incentives to improve their innovation and competition (Johnson and Turner, 2006). This innovation and competitive pressure leads companies to adopt a new technology, which is important for survival (Oster, 1994).

Within the EU, there is a common high level of consumer protection. This framework includes standards about the safety of products that is important for a company to take into consideration (Johnson and Turner, 2006).

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1.5.2 Stakeholder analysis

"*Stakeholder analysis can be used to generate knowledge about the relevant actors so as to understand their behaviour, intentions, interrelations, agendas, interests, and the influence or resources they have brought – or could bring – to bear on decisions-making processes*" (Brugha and Varvasovszky, 2000, p.239).

Through using this analysis, there are seven actors, which are going to have an affect on an organisation. These actors are: costumers, suppliers, government, managers, employees, shareholders and banking institutions (Lynch, 2006). Each of these actors has to be taken into account while implementing new strategies, since either good or bad decisions are going to affect them in different ways. The larger the company, the more the priorities of the stakeholders and the organisational goals may not correspond (Worthington and Britton, 2006).

This analysis shows that there is a link between the stakeholders' interest and what impacts the different parts of the PESTEL analysis have on the organisation. As noticed both external (PESTEL) and internal (stakeholders) factors will have an impact on the decision-making of the company. Furthermore, the linkage between the stakeholder analysis and the sustainability of the environment is that the behaviour of stakeholders will impact on the decisions and strategies of the organisation; hence we are living in a world of globalisation.

2. Methodology

In this chapter the Authors will present the research method and techniques used to collect information about the chosen topic. It will further include the actions taken to create the final dissertation and why other method techniques have not been selected.

Silverman (1993) presents two approaches for researching and studying a topic. The two approaches are based on positivism methodology, (which is a research that discovers the rules for using quantitative methods) and the qualitative methodology, (which deals with hypotheses taken from field research. It emphasises detailed collected data through open-ended questions). These kinds of methodologies and theories cannot be true or false, but only more or less useful.

Jacobsen suggests that the qualitative approach is analysing the depth of the problem, when the data is distinct and sensitive to unexpected circumstances. It concentrates on few aspects to clarify the problem (2002).

The quantitative approach is an approach to find the width of the problem by exploring many units, which are examined only relatively, to a further extent to enable the generalization of the research.

2.1 Position of methodology

Reading of relevant books, articles, journals and Internet reviewing of the authors resulted in a good understanding and clarification of the topic area.

For data collection, the authors have used a deductive strategy for gathering information. The theoretical information which has been thoroughly explored, will be compared to information collected through interviews. The outcome of the examination will inform whether the theoretical information corresponds to the reality or it must be rejected (Jacobsen, 2002).

2.2 The qualitative approach

The qualitative approach includes methods more or less relevant for gathering information about the topic. These methods include first the individual interview, which is considered an open interview, second the group interview, where many people are interviewed at the same time, and the third is that based on observation, which means studying people in different situations, whether they are aware of it or not. In addition to this a document exploration could be used for gathering secondary data (Jacobsen, 2002) that will offer information, which is considered non-quantified data (Silverman, 1993).

The qualitative approach is referred to interpret how people understand a given situation. Moreover, it is suitable for explaining the connection between the individual and context. The approach may well be used when the researchers want to have open questions for unexpected events (Jacobsen, 2002).

For a better analysis of the topic area, the authors decided to use the interview approach to get a deeper understanding of the research. An open interview will maintain an understanding of the subject matter and will further give the ability to ask additional questions. In addition, the subject is much more theoretical than to be generalized in statistics.

The advantages of this approach are that the gathered data through interview will provide more detailed answers and are unique for every interview object. Furthermore, the researcher can gain real understanding of the phenomenon, while the questions are not forced upon the person interviewed because it is flexible. The disadvantages are that the interviews demand a lot of time and deep studying of the subject. This method is complex and the flexibility can cause a problem for the researcher to achieve the final view, since new information appears all the time (Jacobsen, 2002).

2.3 The Alternative – quantitative approach

The quantitative approach is more appropriate when there is much information provided and the problem is relatively clear. Moreover, it is suitable to describe the frequency and extent of the phenomenon. In order to find information about the topic, a questionnaire with given answers is relevant. The method is relatively individualistic and the information is restricted to each individual, not to groups. It is the total sum of the individuals contribution of relevant information, which allows the approach to be generalized; yet, one can say that the context is very limited (Jacobsen, 2002).

The researchers have not selected this approach because they believe that it is not suitable for collecting data for the topic in question. The aim of choosing the relevant method for this area is to get a deeper perspective to increase the understanding of the concepts of “sustainability” and “environment” and see that it could be considered as a competitive advantage for companies. Therefore, the quantitative method cannot be applicable for this study.

2.4 Collection of data

The authors have collected the research data basically by the use of primary and secondary data. The primary data is collected at the first hand by the researchers mainly through interviewing relevant companies, while the secondary data comes from previous studies and is collected through books, articles, journals, Internet sources and case studies (McNeill, 1990).

Collecting information through secondary data is quite simple and cheap, although it is hard to get the relevant information provided for the topic (Molin, 1975). On the contrary, primary data is more difficult, time consuming and expensive, since it has to be collected by the researchers themselves.

2.5 Literature

Relevant literature has been gathered through reading of books, articles, journals and Internet sources. The academic books have been found at the University of Lincoln Library, while the journals and the articles have been collected from databases *FACTIVA*, *ABI/INFORM, ABI/INFORM Global* and *newspapers*.

The authors have composed the theoretical data about sustainable development, environmental issues, strategies, competitive advantage and economic information from the relevant literature. The review is describing, summarising, evaluating and clarifying the studied literature.

2.6 Case Study

A case study is the activities rooted in reality, which can only be studied and understood in context at the given time. It could either be an individual, groups, institutions or a community. Choosing to use only one of these is called a single case study, whereas studying a number of them is called a multiple case study (Gillham, 2000). The authors have chosen a multiple case study, because it can, on the one hand, provide a lot about whether the theory may or may not hold in comparison to the reality and, on the other hand, whether different conditions might affect the study.

Furthermore, the multiple cases are more influential and convincing and are focused more on the depths of the insights than single case studies (de Vaus, 2001).

The case study contains two approaches, which are theory testing and theory building case studies. The authors have selected the theory-building case study, because this approach can help to develop and improve the propositions of the theory to fit the cases that are studied. The theory building approach “begins with only a question and perhaps a basic proposition, looks at real cases and ends up with a more specific theory or set of proposition as a result of examining actual cases” (de Vaus, 2001, p. 223).

Through reading literature the researchers have gained the insight that sustainable development is not completely achieved, which has resulted in the choice of a multiple case study. This choice is based on the background (see section 1.1), whereas the companies can gain more competitive advantage through adopting the environmental sustainability development. Exploring multiple cases will provide more insight to clarify the aim of this research (de Vaus, 2001).

2.6.1 Respondent selection method

When it was time to start the research, the authors found difficulties with selecting the objects of the research. They first wanted to interview big international oil companies such as Shell, Petroleum and CONOCO which are implementing the sustainable development strategy and compare it with the Swedish oil companies. Unfortunately those companies in England did not wish to participate and therefore they have been ignored. The authors then decided to interview some oil companies in Sweden such as Shell, Statoil and Preem. Some of these companies have already implemented the sustainable development strategy and worked with it for a long time to protect the environment, while the others are planning to implement it. The authors have chosen these three companies just to make a comparison between the companies which has implemented the sustainable development strategy and the one which has not implemented it.

We have interviewed the CSR manager and the Senior vice president of Environment at Statoil, the environmental director at Shell as well as the Chief Accountant and the environmental co-ordinator at Preem to get better understanding about our study.

2.6.2 Approach

The number of the respondents was limited to five after our initial preparations of the case study. Through these respondents, the authors wanted to delve deep into the issues they wished to examine. The researchers made contact with Shell's environmental director Per-Olof Lindh, Statoil's Senior vice president of Environment Tor Fjæran, CSR manager Willy Egset, and Preem's environmental co-ordinator Martin Sjöberg, and the Chief Accountant Göran Byström. Even though some of them have different positions they have almost the same point of view about the general information of Sustainability and its advantages and possible disadvantages on the company's economy and the business. In fact, they provided some important information about the research which showed the connection between the three pillars as is showed in figure 2.

First of all the Swedish environmental co-ordinator at Preem was approached in order to understand the concept of sustainable development and its impact on the organisation's economy and competitive advantage, and to perceive the relation between the stakeholders and the sustainability program. Secondly, the Chief Accountant at Preem was contacted in order to examine the economical point of view. Thirdly, the environemental director at Shell contributed with his point of view about all the questions. He had been in contact with a collegue before the interview to achieve the correct answers. Finally, the senior vice president and the CSR Manager at Statoil was approached to get good information about the company's sustainability and the environmental view.

2.6.2.1 Interview

The disposition of our study led to an open-ended interview. This manner of interview is the most relevant method for collecting information within the qualitative approach. The collected information could be seen as only key words, sentences or informants. The open-ended interview takes place between two persons or more than two persons in form of either face to face interview or via telephone. While the interview process is going on there will not be any limit for what the respondent will say. The result of gathering data by interview could look like a quantity of notes which later on must be analysed. The open-ended interview can lead to collecting some individual points of view as well. Furthermore, this form of interview could help to provide good and deep information which helps the researcher to understand the context of the explored subject (Jacobsen, 2002).

The authors have chosen to attend interviews without any guideline. The questions were not organised in any specific order. The way the questions were asked was dependant on the nature of the interview. The relevant information has been collected and some particular areas have also been examined through interviews which were formed as an ongoing dialogue between the interviewer and the respondent. The prewritten interview guide (see appendix 1) had been sent to the respondent before the interview day. The appendix includes the issues which the authors wished to examine. This was to make the interview process easy for both the interviewer and the respondent which helps them to prepare themselves before the interview day (Kvale, 1997). It was first planned not to give the respondents the questions, but they wanted to have a look at the questions; though, other additional subjects have been discussed and the result was to gather wide information about the concerned area.
The authors have chosen to conduct telephone interviews for collecting their primary data since the physical location of the respondents were too far away and they did not have the possibility to conduct face to face interview. The respondents' location at Shell and Preem were in Stockholm, and those of Statoil were in Oslo.

The telephone interviews gave two positive side effects. Firstly the interviewer's impact on the respondent can somehow be decreased and secondly the transport costs can be reduced as well (Jacobsen 2002).

The researchers are considerate about some disadvantages of using telephone interviews and the impracticality of not seeing the interviewees. Moreover, telephone interviews lack the possibility of seeing the respondents' body language which can say a lot and it can also be seen as a casual way of conducting an interview. Some investigations emphasise that it is easier for the interviewee to lie. The authors cannot see how the respondents act or behave during the interview (Jacobsen 2002).

2.6.2.2 Interview guide

The authors used pre-written primary questions (see appendix 2) when conducting the interview approach for their study. The questions were used differently according to the respondents' position within the organisation. The same basic structure has been followed with all the five interviews, but there has been some variation between the interviews which shows that the questions were more open. This result was that the authors did not receive response for some questions and some other questions have been added since the interviews were more like a dialogue.

As Jacobsen states, the interviewees' or organisations' names can be concealed if they want to. Furthermore, the interviewers can often face difficulties with writing down the data during the interview process and, therefore they may ask for recording the interview but the interviewees can refuse it if they do not want that (2002). When the interviews were finished and the primary data was gathered, the result was shown to the respondents to confirm that a particular data had been collected.

2.7 The analysis process

The analysis process has been described by Dey (1993) and Jacobsen (2002); as similar to the shape of a spiral which is rising continually in order to clarify the empirical data into relevant information to help the purpose of the study. This kind of analysis has been used in our study in order to promote the process. A big amount of data was collected through conducting interviews which was categorised according to the different positions and possible points of views of the interviewees within the organisations. The interview process could highlight and clarify some important points in the case study.

The empirical data was categorised once again when the process of the analysis started. The empirical data was categorised this time according to the subject and the points that were discussed. The theory was classified in the same way and this made the process of the data comparison to the theory easier. Finally, the gathered empirical findings were also compared to the theory and the authors could find a combination to the theory. This could perceive a connection between the finding and the theory.

2.8 Evaluation of sources

The qualitative approach needs to be critically evaluated to measure the quality of the research, and distinguish whether the found information is true compared to the reality or has been affected by the method. Therefore, the empirical data has to meet the requirements of reliability and validity. The validity of the data must be relevant and suitable, whereas the reliability of the data must be consistent and truthful (Jacobsen, 2002). Furthermore, Silverman (2004) points out that these issues are relevant, since the aim of the research has to be objective and credible to produce a description of the reality. Through the research in question, the authors strive to fulfil the reliability and validity to measure the right information in the right line of action and minimize problems that might occur (Jacobsen, 2002).

2.8.1 Reliability

Reliability means the explored information must be trustworthy. The research has to be carried out in a truthful way that can convince a reader and meet its expectations (Jacobsen, 2002). In the qualitative approach, the collected primary data will be compared to the secondary data to find if there is a linkage between them or not. The more correlated the measure is, the more reliable is the outcome (de Vaus, 2001).

Kirk and Miller define reliability as “the degree to which the finding is independent of accidental circumstances of the research” (1986, p. 20). Hence, to achieve a good result of the study, the researchers have to be objective; otherwise, the conclusion will fail to deliver a quality image of the reality (Kirk and Miller, 1986).

The authors want to obtain a result which measures their interest of the study and that they can trust the gathered information. As the proverb says “Do it right, or don’t do it at all!” (Jacobsen, 2002).

A high reliability of the interviews has been achieved as the authors have made their best effort possible to arrange the interviews in a structured way and with suitable case studies. The interviews were well prepared and even though they were made on a distance between the interviewers and respondents, it did not affect the end result as there was room for open questions during the interviews. The authors were trying to be as objective as possible while giving the open questions and also tried to find as many incentives as possible to ask the relevant questions. Further, the opportunity to follow up the information was given as a frequent contact with the interviewed objects was established. In order to establish a high reliability both authors were present during the interviews and the use of a speakerphone made it possible for questioning, listening and taking notes at the same time.

The reliability could be lowered, as the sustainable development concept mainly has been seen as a praised strategy. The finding of any negative views has been narrow and could therefore impact on the end result. Though, as it has been positively met from many it could be that the reliability of the concept is high and as the study will show the concept has been positively met from the case study objects and could therefore confirm of high reliability.

2.8.2 Validity

Validity means the researchers try to measure what they really want to be measured and to be relevant. What they measure among some groups must be applicable to all groups. Generally the consistency and truthfulness are divided into two different types: internal and external validity. The internal validity examines whether the measured information is the same as the information the researchers wanted to measure. While the external validity is obtained if a result from a limited area is applicable in other contexts (Jacobsen, 2002).

Although the internal validity may be of high quality to this study, the problem is that it is limited to apply the findings to a more widely context. However, a qualitative approach and case studies do not need a high external validity, but instead an internal validity will be more suitable for this research, because the external validity is more generalised (de Vaus, 2001).

While researching the authors found that the internal validity was high. On the contrary, the findings are hard to generalise while it may show of low external validity. A high external validity was not the goal of the study but to have the highest correlation possible between the concepts of the sustainable development strategy (to be more environmentally friendly) and competitive advantage. Additionally, as the primary and secondary data was examined it confirmed a strong linkage as well, which can verify a high internal validity.

It was important for the authors to measure the relevant information to be able to know that the purpose of the study had been met. Hence, every step was made in line with the others in order to obtain this. While researching the authors had a clear guideline and tried to focus on the significance of the study and to stay within the boundaries.

2.8.3 Evaluation of Respondent Selection

Though the selection of respondents for the study have somewhat been done through a so called “snowball effect”, (Jacobsen, 2002), the authors believe that this did not have any affect on the end result of the interviews. However, the researchers could get interesting and important data for the purpose of the study and could match the wish that the authors wanted to discover on sustainable development strategy within the organisation and its competitiveness.

The authors have chosen the respondents within the mentioned organisations which could have some effect on the result. The chosen selection method could give to some extend some interesting primary data which could positively serve the reliability of the study.

In the beginning, the authors wished to interview more than three oil companies. Anyhow, the chosen three companies could perceive very good and useful information on the sustainability process and the environmental effect on such companies.

3. Theory / Literature review

In this section the authors has delved deeper into the central subjects of discussion and in order to find similarities and differences this has been made in a separation of the expressions sustainable development and competitive advantage. The reason of this is to learn more about each subject to be able to find the linkage between them.

3.1 Sustainable Development

In the 1980s, the idea of pollution prevention and protection of the environment emerged. The governments and the industries became more aware of the impact that disposals had on air and water. Therefore, the governments and the industries realized that by implementing a strategy, which could control the waste and the pollution, would gain environmental sustainability and provide a clean nature for the future generations. Basically the misallocation of materials, labour and energy cost companies lots of money, competitive advantage and time. Through protecting the environment and preventing pollution, companies would be able to make more profit, minimise risks, cutting disposal cost, gain competitive advantage and at the same time provide costumers higher quality products (see figure 1) (DiPeso, 2000). Furthermore, companies' responsibility is divided between economic growth, social and environment aspects to be sustainable. Environmental sustainability will provide more opportunities for business and enable them to be more efficient through new technologies (European Commission, no date).

The concept of sustainable development is a wide range of meaning which is seen from different points of view by businesses, governments, environmentalists and social reformers. The sustainable development has become more important throughout the world and today about 95% of large companies in Europe and USA believe in it. It is widely discussed in many organisations such as the World Economic Forum. Further, over 150 of the world's biggest companies in different sectors such as oil and gas, banking and finance, chemicals, logging, autos, etc. are members of the World Business Council of Sustainable Development (WBCSD).

The sustainable development seems to be more supported now among different political parties in the world. Environmental activists as Greenpeace and Friends of the Earth agree with the sustainable development strategy, but they are critical to companies who are members of the WBCSD. Those people and organisations who are supporting the social issues and the sustainable development strategy are disagreeing with the outlook of businesses and international economic organisations. The Real World Coalition believes that the impact of globalisation will not eliminate the poverty, but instead increase it. On the one hand, the sustainable development is a good practice because it is like 'motherhood and apple pie' which sounds very good for everyone who agree with it and is a strength point for this strategy. On the other hand, some argues that it is a meaningless concept which lacks any clear validation analysis or theoretical framework. This concept can mean anything that anyone wants. Therefore, to avoid any possible conflicts between the economic growth, the environment and the social equity, it is argued that it is better to change the word sustainable development to sustainability or sustainable livelihood. Brundtland claims that it is better to concentrate on human needs and the environment, than on these phrases. However, this concept is rejected by Deep Ecologists as it prioritizes the needs of humans and anyhow it is defined to be viewed by the environment from a human standpoint. Despite all these problems the phrase sustainable development has been used as it attempts to hold close the relationship between the socio-economic and environmental view. Therefore, it is clear from all the debates that the concept of sustainable development does not have a common philosophy (Giddings *et al.*, 2002).

Despite this, with an implementation of a sustainable development strategy companies would be able to gain different stakeholders' satisfaction and to be friendly to environmental or "green" issues. However, some companies do not know how to adapt to these "green" challenges, since they are not aware of how it would affect their life-cycle (Karna *et. al*, 2003).

As a result, the sustainable development requires a shift in how the humans need to see the world as they are part of a web that is called the environment and society. The long-term meaning of sustainable development will be integrated and is based on human life and the world we live in (Giddings *et al.*, 2002).

According to the European Commission, policies are not good enough to reach the goals of a sustainable development and a more environmentally friendly world. The responsibility lies in the hand of the society to change the way of living. Society is not only you and me, but also organisations and companies as they have greater impact on the world as a whole. To reach the goals set by the European Council of June 2006 requires a changed vision in which people are thinking beyond their wants and needs and instead focuses on what the future will bring (European Commission, 2007a).

Further on, the authors will discuss the three different objectives of sustainable development since they affect the companies in different ways.

3.1.1 The economic view

In the past decade the prospects of increasing economic growth was the base for the social security system, the health service, social services and public housing. Therefore the economic growth has always been an important issue in order to pay for the welfare state through providing the resources for social policy and redistributing the money from some significant sections of the population. This was used as an opportunity by the government not to increase the taxation as it would have been disliked by people.

In other words the economic view is mainly about growth as is measured through Gross Domestic Product (GDP). Currently it is more accepted as a subject which consider the issues that allocate limited resources. Mawhinney suggests that the economic view and the sustainable development are the same when it concerns the limited resources. The economic sector is seeking the opportunities to use these limited resources, whereas the sustainable development is seeking the opportunities not to use it or to use it in longer term. It is just the matter of seeing it from different points of view. The economic growth is good for the companies and additionally it contributes to the countries GDP. Furthermore it can be linked to the investments in education, health and attitudes to environmental protection. These factors are seen to be important to the definition of sustainable development and benefit all three views. The economists believe that the environmental dimensions can be reduced not only through implementation of the Sustainable Development Strategy, but also in other manners such as economic growth which lead to environmental benefits (2002).

It is argued that there is a relationship between economic growth and environmental quality. As a country's income per capita increases, peoples' demand for a cleaner environment will increase as well. Furthermore, it is argued that the country's economic growth will decline if the society and the government do not take the environmental issues into consideration. Therefore, to save the environment, the economy has to grow on a sustainable base (Panayotou, 2000).

To be able to adjust to the market and its demands is of great importance for businesses. In fact, the importance of producing is not only found in the way the products are being produced or how the waste is handled, but in taking into consideration the whole life-cycle of the production process. The life-cycle-analysis should describe and look into relevant environmental effects such as the handling of raw materials, production, trade, use, recycling, waste handling and transportation between each stage. The life-cycle procedure is both expensive and complicated, but it is really necessary in order to meet the market expectations. A link has been acknowledged between economic growth and change of demand. When people gain more, they demand for better products. The better a company can adjust to the responsibility of the product life-cycle, the better the customer satisfaction. Furthermore, the companies need to demand the same environmental standard of their suppliers and entrepreneurs when dealing with raw materials, transportation etc (Industriförbundet, 1998).

An economic barrier which could appear is the inability of companies to recognise concealed costs of waste, such as treatment and disposal charges, insurance and potential future environmental legal responsibility. It was found by the European Community and declared in the Paris Summit in 1972 that "economic expansion is not an end in itself – it should result in an improvement in the quality of life as well as the standard of living" (Barnes and Barnes, 1999, p. 28).

It is seen that the basic goal of the companies is not only to obtain the required quantities of oil and dominate the market to achieve the highest rate of profit and consequently achieve success in business; but sustainable development is one of the primary goals as well. In achieving acceptable profit margin, the cost cutting is as important as the production volume is. This change has been required by a change in reality because the demand of oil is flat, the

prices are weak and there is a great awareness among people that the so-called oil glut is not a temporary phenomenon. Therefore, the greatest challenge of many companies today is to increase the income and cut down costs.
The technology is changing the way of working in many companies and it's changing the marketplace. It is making the whole world, one big market.

Modern technology is changing the way of working in many companies and the marketplace as a whole. It is making the whole world, one big market, and its impact on business and commerce is prominent. The role of computers and internet is dramatically increasing in all kinds of business, espacially in the industry of oil and oil-marketing. Internet has made business productivity and creating new and easier access to worldwide markets easier, smoother and quicker. However, many businesses see the environmental challenge as a threat and fail to realise the opportunities for developing new markets, new products and new processes. In the short term, there is a need to look at the business environmental impacts and try to find ways in which these can be made acceptable (business responsibility).

The sustainable development strategy helps companies to develop opportunities and manage economic, environmental and social risks. Many investors consider it as a definitive value for success (Cheney, 2004, p. 14; Hart and Milstein, 2003, p. 57).

There may be reluctance within a company, while changing or rearranging the strategic decisions. Finding a way to implement the change into the daily agenda might be a step to overcome such a barrier.

There are some criteria which the companies must fulfill. The criteria are imposed in three areas: economic, environmental and social. These criteria have an effect on the economic-financial management of the companies which can be seen clearly in the accounting indicators (M. Victoria Lopez et la, 2007).

The industries can cope with strong legislation, but only with world-wide regulation. All companies should compete/act in a responsible way in which they can gain competitive advantage from sustainable development without being held back by legislation. Hence, the industries will be forced to limit their use of resources (Industriförbundet, 1998).

Laws and regulations:

There is an international standard system for environmental protection and guidance. The ISO-standard is divided into two parts, the organisation and the products. There are a total of 18 standards of which 6 are completed. When these standards are fully completed, the demand for a sustainable development will be met. The most basic of the standards is the ISO 14001, which deals with the shaping of an environmental policy, the mapping of environmental effects, goals, organisation, education, communication, documentation etc. The European Union has come up with a complementing standard, the EMAS-system (Eco-Management and Audit Scheme), which deals with environmental controlling, how to evaluate this and the importance of informing the public about the progress (Industriförbundet, 1998).

3.1.2 The environmental view

During the 1970s, the environmental debates gained a new status on the political agenda of all governments. The recognition of problems of environmental degradation appeared from the increased use of chemicals. Several member states of the European Union started to control the growing pollution problems from the different natural resources with national policies. Although, the question of this danger grew that it might hinder the world wide trade by these measures. Therefore, the European Union launched the first part of the environmental action program (EAP) in 1972 in order to ensure that new barriers to trade would not appear. This was the key concern of that policy and other concerns was also met. This legislation was including the protection of water and air pollution, waste management and the protection of wildlife and human health, etc. The introduction of EAP has resulted in success as it has had a positive impact on the adoption of legislative acts in the European Union (Barnes and Barnes, 1999).

During the period 1960-2000, the awareness of environmental protection increased among people globally because of the previous discussed issues to remove the worst environmental problems in the developing countries of the world. The acceleration of growth and the speed of change, caused by globalisation, have resulted in worries among people that an environmental disaster may be just around the corner. The environmental issues must be more focused on before the studying of any political and science-based standpoints since it has been the main debating subject among the environmentalists.

Many in the green movement point that there are economic systems, unrestricted desire to control the nature and to leave the markets free without thinking about the consequences as the fault that needs improvement in the sustainable development policy.

The environment has always been an important issue for the humanity. Even in the past, people have always had a desire to control the nature to avoid using an abundance of natural resources and harness for the good of humanity. Furthermore there is much evidence of the impact of environmental damage which is based on scientific facts. This has resulted in the advantages, from the environmentalists' view, that assumptions can be avoided and of the importance of acting in line with a sustainable development strategy. The most important priorities at this time should be to handle the effects of climate change, but also how to handle disposals and toxic materials. Some environmentalists argue that industries should pay the price of failures to handle these as they are often part of the damage, though others argue that industries are the once moving the sustainability process forward as they are developing more sustainable alternatives. The oil companies are part of an industry which is seen as a source of much pollution and which lacks of straightforward decision-making. Nevertheless the companies have started to look at their operations and are trying to find incentives to promote alternative products which are more environmentally friendly (Mawhinney, 2002).

3.1.3 The social view

The sustainable development has often been divided into economic, social and environmental, and it is trying to bring these three categories together in a balanced way as it is shown in (figure 2), avoiding conflicts between them as well (Giddings et al, 2002).

Social sustainability also includes the system of social organisation which lightens poverty. In other words, social sustainability establishes the bases between social condition such as poverty and the environmental decline. Even its history explains a negative linkage between organisation dominate, the level of poverty and the use of natural resources. Some argue that the environmental sustainability is an essential and important step for economic growth and poverty alleviation, or the economic growth and the poverty sustainability are more needed and should have a higher priority than the environmental sustainability which can be addressed.

According to Steurer et al, Brundtland's report concludes that "a strong economy is a prerequisite rather than a burden on healthy environment" when it is concerning economic growth issues (2005, p. 264). While there is some evidence which states that the environmental sustainability may be more necessary and prepare a good condition for better economic growth and poverty sustainability. The theory discusses that the social equity requirements hinder the economic growth which is considered as an important element for the social sustainability. In order to link these two categories together, the environment must be able to create an optimised use of resources, prioritise resource allocation and foster equitable resource distribution (Basiago, 1998).

Basically, the sustainable development strategy is a well known social guiding model which integrates the economic, social and environmental issues in all levels in the short and long terms. Actually, this strategy ought to be followed by everybody in a different way. The governments contribute to the sustainable development policy and it is often involving some sorts of regulatory force, which the managements of companies should more or less willingly apply. With the term "more or less willingly", stakeholders influence come into play and they can affect the decision making in any company. Stakeholders (including governments) are important for transmitting the sustainable development strategy from society groups to the world business (Steurer et al, 2005).

3.2 Competitive advantage

"A competitive advantage is said to be sustainable if it cannot be copied or eroded by actions of rivals, and is not made redundant by environment developments" (Wit and Meyer, 2001, p. 201).

The adoption of a sustainable development strategy has become more important for companies as they can not only be environmentally friendly but also gain a competitive advantage. Though, to be competitive the companies need to find the edge in how to control the pollution, and to learn continuously, transform and be innovative in order to reduce the environmental impact, create social value, and to go beyond its competitors.

Companies are often facing the problem of a gap between what they want to do and what is the correct thing to do. Hence, by diminishing the gap people's anxiety will be reduced and it will further lead to customer satisfaction, which is a large part of competitive advantage (Papmehl, 2005).

Competitiveness could be achieved either through cost leadership or differentiation. Cost leadership means to produce products at a low cost, but can also mean to produce with fewer spills of materials. Differentiation on the other hand, means to be innovative and provide products which differ from the competitors. Porter suggests that "differentiation can be based on the product itself, the delivery system by which it is sold, the marketing approach, and a broad range of other factors" (1985, p. 14).

The most important way to achieve competitive advantage is through service customers. Actually, very few companies in the past had a clear definition and useful strategy for customer service. This state has changed rapidly more and more as companies has started to recognise the importance of offering a good quality and a good customer service for achieving differentiation and competitiveness over competitors. Through paying attention to the service delivery system and service quality companies can gain competitive advantage. The service quality can be achieved through control of the delivery process which is the same way in which product quality is achieved through careful control of the product process.

When we talk about a delivery process it refers to the service encounters which affect the customer's satisfaction. It is proved that in some companies, the profitability and competitiveness has increased through customer satisfaction and retention. Today many companies try to maintain customer satisfaction and loyalty (Payne *et al.*, 1998).

Furthermore, the firms are looking for the competitve advantage in the first class that operates in a complex global environment, in order to certify the capacity to create value in the long term. Currently, the advantages are linked to the adoption of social responsible behaviour. The interest has been shown in those issues where the emergence of sustainability indexes are linked to the financial markets.

By gaining such advantages, the companies would be able to survive and obtain an acceptable profit rate and economic equilibrium. It is argued that those companies which have adopted the sustainable development strategy grant competitve advantage over those companies which have not such strategy (Adams and Zutshi, 2004, p. 34; King, 2002).
The information on sustainable development strategy clearify that the company which develops and discloses this strategy should simplify the development of better systems of internal control, decision making and cost saving (Adams, 2002). The efficient management can develop capabilities that enable long term competitve advantages easier. The study of the framework of sustainable development strategy and adopting it can create value; where value creation refers both to achieving acceptable profit and satisfying the demand of stakeholders.

In the last decade, the society started to demand that companies must use policies that move towards sustainable development. The philosophy of sustainable developmnt strategy assumes that we desist from a narrow version of classical economic theory and develop corporate strategies which contain goals to maximize stakeholders' interests.
The companies can contribute to sustainable development by re-establishing their operation and processes. This can lead to that the companies will provide economic results which are sufficient to enable the business's viability but the company's first concern must be its survival.
The management of environmental, natural, economic, social and political factors has provided the sustainable development strategy. It is thought that this strategy creates value, and furthermore, it must influence the accounting indicators, if we are able to talk about the

impact on performance; that is if the adopting of sustainability strategy contains changes in performance (M. Victoria Lopez, et. La., 2007).

A business market appears when customers demand for more environmentally friendly products,which will further result in a business opportunity, when businesses realise the competitive advantage that could be reaped. Unless there is a suction power from a paying market or a pressure power from laws and regulations or competition, businesses will continue as usual. The stronger the competition is in a market, the more efficient the development will be. Moreover, the market will grow faster, which is a strong force for companies to seek for opportunities of competitive advantage (Industriförbundet, 1998).

It is of great importance for companies to develop a USP ("unique selling proposition") to offer to its stakeholders. The USP is about letting pressure groups know what the business is really about. It could be about product development, investments and even mistakes of which the company has corrected. The most important thing is letting others know the uniqueness of the company, what the company stands for and what the future will bring about and it should be communicated in an easily understandable way (Stimmel, 2007).

A new strategy which could be used to gain competitive advantage is to seek opportunities with a reverse chain activity. This means the company reuses, recycles etc. the products which its customers can return. Researches show that customers return about 6% of used products and with knowledge about what companies can do with returned products, a loyalty can be built between company and customer (Jayaraman and Luo, 2007). For oil companies this could be a new era of environmental protection as the handling of waste would be secured.

4. Empirical Data

The empirical data has been collected by interviewing some particular individuals within the selected companies (Shell, Statoil and Preem). They have had certain knowledge about the sustainable development strategy and the environmental view. In this chapter the authors will describe the objective of the case study, the respondents and the empirical findings of the interviews as well.

4.1 Case study objects – Shell, Statoil and Preem

In this section the authors will present briefly the background of the three interviewed companies in order to facilitate the process of comperation between the empirical data and the theoretical data.

4.1.1 Shell

Since 1907, the Shell Group has grown and developed to be an oil company with activity in over 140 countries and with about 112 000 employees. The company is not only known for their gas stations but also for their transportation and commerce of gas and oil, fuel production for airplanes and ships, marketing of natural gas etc. Furthermore, the Shell Group is investing in renewable energy and is trying to find incentives to produce energy with less coal which could award them with a competitive advantage.

The company is actively working in line with industries, governments and social networks in order to deliver what is expected of them in social, economical and environmental questions (Svenska Shell, 2006).

Shell was established in Sweden in 1912 and is today operating about 385 gas stations (Svenska Shell, no date).

In 1997 the Shell Group implemented a Sustainable Development strategy after pressure from the society, stakeholders and governments etc. The company wants to be ahead of their competitors and have therefore outlined a policy to be the first to come up with new evolutionary products and solutions to the environmental problems. The company wants to have an open dialogue with their stakeholders about both positive and negative impacts and to identify business opportunities in the future (Shell chemical Ltd, 2007).

4.1.2 Statoil

The Norwegian company Statoil was established in 1972, is operating in 34 countries worldwide and is having production and exploration activities in 15 of these. The company is employing over 25 000 people. Statoil is an efficient producer and transporters of both oil and gas and one of the world's major retailers of crude oil. Furthermore the company is a significant supplier of natural gas to the European market (Statoil ASA, 2007a).

In 1985 Statoil acquired the oil company Esso and was established in Sweden. Currently the company is operating over 600 gas stations there. The wholly owned subsidiary, Svenska Statoil AB sells, distributes and markets diesel, heating oil, wood pellets, aviation fuels, lubricants and LPG (Statoil ASA, 2007b).

The company implemented a Sustainable Development strategy a few years ago and has been acknowledged for their environmental work. Statoil's main goal is to "create value for their owners through profitable and safe operations and sustainable business development without causing harm to people or the environment" (Statoil ASA, 2007a).

4.1.3 Preem

Preem was established in Sweden 1996 and has grown to become Sweden's largest oil company with over 500 gas stations and 4000 employees. The two oil refineries which are owned by the Preem Petroleum AB are known to be the most modern and environmental- and energy efficient in Europe. Preem is on the one hand selling gas, diesel, oil etc on the Swedish market and is on the other hand selling petroleum and processed crude oil on an export market mainly to the northwest of Europe (Preem, no date a).

Even though Preem has not yet implemented a Sustainable Development strategy the company is working towards the main pillars of it and has recently started to outline their own strategy. It is important for Preem to listen to the stakeholders and to be open about the impact which the company has on the environment. Some of the things which the company is striving to obtain are to lower the environmental and social impact of the products both globally and locally while being transported, produced and/or used (Preem, no date b).

4.2 Primary Data

The primary data was collected through interviews of substantial persons within the three oil companies Shell, Statoil and Preem and will therefore be reported separately in order to be analysed correctly in the next section.

4.2.1 Shell

According to the company's environmental director, Per-Olof Lindh, Shell committed to contribute to the sustainable development in 1997 and has subsequently made both big and small actions in line with the company's business strategy. It has required a balancing of short and long term interest. Furthermore, an integration of economic, environmental and social considerations has been implemented in the decision making. Lindh state that Shell's view of Sustainable Development is the one of the Brundtland Report:

> *"Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs."* (Wikipedia, 2007)

The idea is to provide a better quality of life for everyone, now and in the future and at the same time achieve a stable economic environment. Shell has chosen this definition as many of their stakeholders has the same view.

Since the introduction of the new way of acting Shell has been trying to influence and be influenced through attending to meetings with WWF, Amnesty, Greenpeace and political parties. Lindh says it contributes to both the company and the organisations as it helps them to set goals together. However, he thinks it is difficult to see whether the introduction of SD has changed the company or not, but claims that the actions taken are part of the worldwide questions.

Concerning the stakeholders, Lindh claims that they have power of the decision-making in the company. For example, the society and the big businesses are working in line to become more environmentally friendly, but also the government need to set boundaries for the company to work against. He mentions that frameworks make it easier for each company (and the society) to know that the goals are reached. He further claims that it is important with policies that reward the companies in order for each company to be treated equally in terms of actions.

As to the economical part, Lindh claims it is difficult to know whether the commitment to sustainable development is being an issue to the company, instead he think it brings advantages as it would not be possible not to implement sustainable development in the long run. As the concept is an integrated part of the budget many of the actions taken are dependent on it. In year 2006 Shell increased their spending on R&D with 50% and much of the R&D is spent on finding, producing and refining the oil and renewable energy. The company's cost is mainly from handling the disposals and an example of this is that of a leakage in year 2003, in which Shell was fined to pay 1.5 million Swedish kronor.

The main concerns that Shell is focusing on is firstly, the handle of waste. The company's vision is to have no waste at all, but today it is problematic to handle the different products. Therefore the company has built a secondary containment in which the oil tanks are built-in so that the depots where the products are handled are safer and it has proved to reduce the spill. According to Lindh, the cost of this has been high but, he claims it will bring competitive advantage to the company. Secondly, the focus is on the products.

For Shell it is important with products which are compounded of as harmless materials as possible. Sweden, for example, was the first country to introduce the sulphur-free diesel in 1992 and Shell was the first company. It took 2-3 years for the competitors to enter the market.

The answer to why not all countries have introduced this product Lindh explains it is a matter of high costs and political impact. Thirdly, the focus lies on the climate change. The company has decided not to sell any light fuel oil, since it has no future. Instead they focus on biofuels. When it comes to fuel, Shell is working towards a goal in which biofuel with 10%-15% of ethanol (today it is only 5%) should be offered at every gas station within a few years time.

According to Lindh, these concerns might change in line with problems appearing along the way, but as it seems now the main problems are outlined and the company is working towards the set goals for these. He further claims that it is important for Shell to be environmentally friendly, since on the one hand, the society is the ones with interest in the company and their influence means the company can go on and on the other hand, a clean environment will provide a better living for people, which is an important aspect of the sustainable development strategy of the company.

Lindh claims that the society has many possibilities to impact on the company especially when establishing abroad. It is then important to create a trust for the company and listen to different stakeholder views of how to handle each step. As seen in figure 4, the company's operations and performance are based on different impacts.

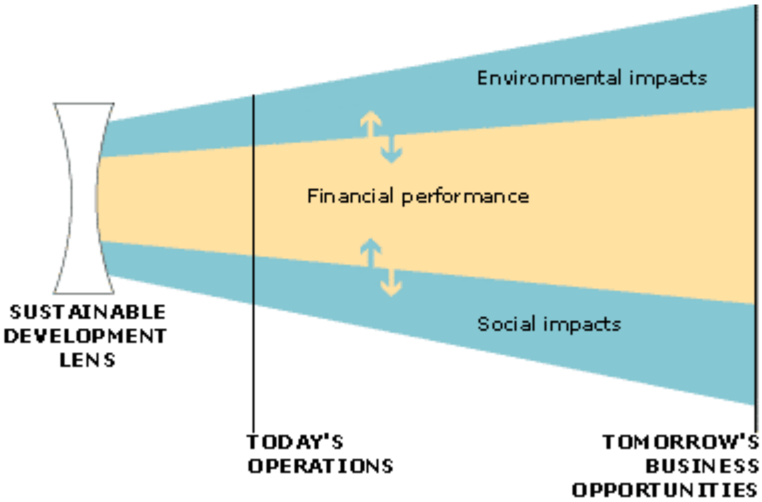

Figure 4, The sustainable development lens, (Shell chemical Ltd, 2007)

In order to achieve a sustainability Shell is investing in renewable energy, mainly in wind craft. Lindh says the company has interest in other companies in order to develop this. Furthermore, partnerships have been established. One partnership is Choren and the interest lies in synthetic diesel.

According to Lindh, a sustainable development strategy is adding value to the company as it is a stable base on which the company rely on to become more responsible towards people and their needs. Lindh says people will always depend on the oil companies to deliver what is expected of them in a responsible way. The oil market is going to increase in the future and Shell would like to be integrated in the development of it. Further, Lindh claims that Shell is being responsible and he thinks Shell has come a long way ahead their competitors in the sustainability, especially in the diesel fuel question where the company is leading the development. He also claims that the company is trying to maintain a good reputation in Sweden as well as internationally.

Lindh also think sustainable development is helping companies to gain competitive advantage as it triggers them to develop a good reputation to the customers. It is important to stay sustainable in order to maintain costumers' satisfaction.

They are developing new products constantly through reinvesting most of their profits that they gain so that these new products will meet the future energy needs, to improve safety and environmental performance. Furthurmore, they concentrate on developing new energy technologies. They have increased their spending on research and development by 50% last year in order to be able to provide the customers with a cleaner, safer, and more sustainable world and supply chain relations which match the business that can gain competitive advantage. It is obvious that they are minimizing chimical and toxicologies from the products to ensure that they are more environmental friendly. These products are petrol, diesel and the technology which has been developed in the Shell group.

It is important to increase and maintain the supplying of modern enegry because it is vital for economic and social development.
Shell is about to reduce their costs through ecco-efficiency, which is about producing with less energy and materials. They can minimise waste and even turn the waste in stable products by adopting new cleaner technologies, recycling and reducing emissions. These activities help them to improve the efficiency of their operations and reduce costs. Furthermore, it avoids costs of the curent and the future emmissions and even create new incomes.

Last we asked if Lindh wanted to add anything to the study and he then responded that Shell is working as a part of a global network. The company is following the policy which the international Shell group has established and he is satisfied with what has been accomplished.

4.2.2 Statoil

The senior vice president of environment at Statoil, Tor Fjæran, says the company's sustainable development strategy is influenced by the three aspects economy, society and environment. Statoil's definition of the concept is about contributing to a sustainable development through creating positive economic result; develop the society and not harming the environment. He adds that this is the way Statoil deals with sustainable development in all the company's activities.

In order for an involvement and development of the sustainable development strategy the company has to work for a positive value of the three factors economy, society and environment. The different processes in which Statoil involve sustainability in their work differs from project to project and need to be taken into consideration in each case.

Ever since the introduction there has been a continuous and gradual development of the company and much understanding has been learned of the practical implementation of sustainable development measures. Further, Fjæran points out that he thinks this has been the case for all companies which have implemented a sustainability strategy. At present Statoil is learning a lot from the company's increasing international activity.

According to Fjæran, Statoil has received mostly positive feedback from the company's stakeholders. The company has been number one on the Dow Jones Sustainability Index for integrated oil and gas companies for three years in a row. He sees this as a positive feedback for what the company has been trying to accomplish.
Fjæran says a continuous involvement with a variety of stakeholders affects the decision-making, design and technological solutions. For example, Statoil had an early dialogue with different fisheries which influenced the way the company developed the project Snøhvit in the Barents Sea.

The environmental sustainability policy has a strong impact in all phases of the activity in the company. Fjæran explains the most important issues for Statoil to be the climate change mitigations, water resource management, oil spill response, discharges to sea and waste management. He continues to say that he thinks the focus will change in the future but all aspects will be important to take into consideration. At present the climate issues are the dominant aspects globally.

Fjæran claims there was no specific pressure related to the implementation of the sustainable development strategy but he feels there will always be a pressure from stakeholders. Further on, Fjæran believes the company has to stay environmentally friendly wherever they operate. It is a must for the oil and gas industry and he thinks it brings competitive advantage to the company to be ahead in the aspects of the environment.

Willy Egset, CSR (corporate social responsible) Manager at Statoil, means any company exists in societies, both local and global, and therefore will always be influenced by it in every operation the company makes. The society defines all Statoil's operating parameters except the hydrocarbon itself.

The driving force behind sustainable development in Statoil is the rapid change, in other words globalisation. The power of emerging economies and resource holders relative to international companies has strongly increased. Further, expectations and requirements of home governments have also increased. This has proved to be the business case for social sustainability.

Egset continues and claims that the company faces the same problem now as it did before implementing the strategy as there is no single "sustainable development strategy" but the strategy is engaging the society and evolves daily. He says Statoil has come up with own CSR (corporate social responsible) and environment policies. He adds there will always be issues, everyday and in every country.

According to Egset, Statoil takes a number of actions to achieve the sustainability policy. For example, in the social arena the actions are materialized in the CSR country plans, and in addition management and staff engage in strategies everyday to deal with sustainability. He further illustrates the importance of a sustainable development strategy to be inevitable. No company would survive without taking sustainable development into serious consideration. For Statoil the work for a functioning strategy has resulted in a number one place on the Dow Jones Sustainability Index for integrated oil and gas companies (as mentioned earlier). Egset thinks the company can put endless effort into developing the sustainability policy.

Egset explains the contribution to environmental sustainability comes from recognition of responsibility and its long term business interest. Therefore the concept should not be used as a competition between the companies but as a framework to create a better future environment. Though, being sustainable to the environment can gain competitive advantage to the company for a number of reasons as for example being number one in contributing to

the development. Statoil will always try to be sustainable and Egset says the reason for this is the company will go out of business without the concept. As the customers' satisfaction is of great significance for the company it will be inevitable to work for a better society and environment.

Statoil is trying to improve their products through using a project in order to achieve an average recovery factor of 55 per cent from sub sea-completed fields managed by the group in 2008. This project is called sub-sea improved oil recovery. In this case new technology will be used to increase the average of recovery factor for mature fields while contributing to good solutions for new projects. These development activities cover cost-effect drainage points.
Statoil is investing heavily in improving their products quality.

The company's goal is to increase their earning in the future compared with the 2004 level, to improve the growth of their output faster and to reduce the cost of the production. At the same time, the target is to deliver this production increase with the competitive profitability.

Statoil has an ambition to increase its own production of gas from today's level. Therefore, a number of new contracts have been negotiated. These developments created new business opportunities for Statoil and at the same time it creates added value. Furthermore, the company will seek to supply Europe from several sources in the long term.

Statoil's target is to create value for their owner through profitability and sustainable business development without causing harm to the people or the environment.
One of their financial objectives is to ensure long term value creation. Furthermore, the greatest challenge in long term value is to ensure the balance between the profitability and production growth.

4.2.3 Preem

The environmental co-ordinator at Preem, Martin Sjöberg, explains that Preem has not yet implemented a sustainable development strategy but has been introduced to it and is working towards one. The company's ambition is to be able to contribute to future wants and needs and to be part of the future development of the oil business.

Sjöberg claims that Preem is trying to implement sustainability into the company, while making their business reports and is going to release a sustainable development report within a few years. Today the company's systems and text work are not telling enough about sustainability, but Preem is working in a sustainable way. The company has made some investments in 15-20 years and Sjöberg points out the importance of sustainability as without it the company would not get any payback.

Preem is rarely getting any feedback from their main stakeholders but according to Sjöberg politicians, researchers and universities are giving some, since they make sequent studies of different subjects. Further on, Sjöberg says the stakeholders are very important in the decision-making of the company. For example, the customers are influencing with their demand, the competitors are affecting how Preem is acting as the company want to be first with new products and further the society and politicians are affecting what decisions the company takes.

According to the Chief Accountant Göran Byström, Preem is poor in the information about sustainability. Therefore it is hard for him to comment about the work but he understands the importance of the concept and can still answer some of the questions made. He claims there is no tax on disposals but the company could be faced with fines if they handle them incorrect. Preem is not having a R&D budget but are constantly developing their products. Byström declare that the company's competitive advantage is to provide their customers with less harmful products of all the Swedish oil companies. He further claims that the more environmentally friendly products, the more the company can sell as both society and authorities are demanding clean products. Preem recently build a refinery building for 3 billion Swedish kronor in order to become more environmentally friendly.

Martin Sjöberg is explaining the environmental policy to be an impact of sustainable development. Both internal requirements and guidelines are making the vision of the company. About 80% of the work done is outlined from the policy. For example, the company is constantly developing the crude oil and other substitutes in order to cut it down to zero use in the future. Further Preem is making progress in their refineries. All these visionaries are controlled by the company's stakeholders.

According to Sjöberg, the main environmental concern is that of carbon dioxide as it is a very actual question in our time. It has proved to have a negative impact on the environment and have to be taken into consideration for a clean future environment. As the climate is the most important query at the moment and in the future he thinks there is a need for a continuous effort on the longer run to handle the impacts.

Preem is facing a pressure to have an outlined sustainability strategy as the company think it will bring a more entirety to it. The strategy would mainly be about how to change the society's oil demand and at the same time increase the welfare. Sjöberg agrees that it is difficult to find a vision to be able to implement this but it is in Preem's interest to find the incentives to do so. The company will never be completely environmentally friendly but is trying to maintain a strategy where they can be as harmless as possible to the environment.

Sjöberg says the society has a total impact on the company. Preem is ruled by partly goals set by the European Union and partly from Swedish goals of the society. He further claims that Preem wants to be part of the achievement of these goals. The company is willing to be active in order to attain and contribute to the development of environmental goals.

As Preem does not have a strategy for sustainable development the main drive is to complete one as Sjöberg think it will make the company attractive and interesting to the society. This will further contribute to the company's long-term gaining. As mentioned a drive is to be active in the influence of measures of development. Sjöberg points out that Preem would have had a sustainability strategy a long time ago if the company would have been listed on the stock-market. However, the company is ruled by a private person.

According to Sjöberg a sustainable development strategy would add value to the company in PR gaining's. He further thinks it would increase the profit and competitive advantage since it would allow some companies to become market leaders. Sjöberg means the company can push the sustainability development as far as possible. As long as there is demand from the society Preem will be able to develop in the future.

Sjöberg points out the company's active social work and participation as a competitive advantage. Even though Preem does not have an entirely designed sustainability concept the company can still gain competitive advantage but Sjöberg means it would gain more with a complete strategy. Today Preem is trying to co-operate with other companies but in order to have a competitive advantage Preem need to be first with new products and to be leaders of the questions. He explains that the company benefits of what is achieved even though they would gain even more with a complete strategy.

Preem puts society at first place and the customer at second and Sjöberg says the customers are of great importance since the company benefit of what the customers buy but as the society needs to be clean for future generations it will always require to be put at first place. Though, he does not se anything negative in this as the customers are aiming for the same goal.

Preem is constantly working with product development and one of the company's goals is to replace domestic fuel oil with ground heat and distant heating within Sweden. At the European level, the fuel oil will be replaced by gas. Furthermore, the company is developing its production of non-fossil fuel. The ambition is to blend in green hydrogen gas of which the fuel will contain 10% within 5 years in comparison with the 5% at this moment.

Operating at the Swedish market is an advantage according to Preem. An argument for this is the strong but fair demands from the Swedish government and other stakeholders. The company is constantly trying to improve its actions and compete with other oil companies in order to improve the market. Additionally, Preem is working closely with stakeholders as for example governments to develop the product life-cycle in line with governmental policies. The cost is somewhat high but it is seen as an investment instead of a boundary by the company.

The responsibility of delivering the petrol has been put on a partnership. Preem is controlling the course of events but is relying on the partnerships promise of health, environment and security policy. Moreover, the cooperation with local retailers has been established. The light fuel oil and other products to smaller companies are being carried out by these local retailers in order for Preem to focus on the development of other business opportunities. Furthermore, Preem is distributing its own fuels with the requirement of double layers on the hull of the ships. Moreover, the transportation is being controlled and coordinated. At this point, Preem's ambition is to seek business opportunities with their distribution.

Preem's refinery is certificated according to ISO 14001. The European Union's goal of 2006 was that transportation fuel would contain 2% of renewable resources and the result of Preem fuel was containing 2.3%. Moreover, the European Union goal is to increase the number to 5.75% until 2010. Though, Preem's ambition is to increase it to 10% but this would require changed European regulations.

What Preem has noticed in the last few years is a growing competition on the market. The fight about the prices has grown rapidly and led to low profitability for the whole line of business. As a result the companies have been trying to find ways of gaining competitive advantage through product development. Additionally, Preem has realised the opportunity of exporting fuels and the profits which could be reaped.

Sjöberg completes the interview by adding that the companies are important in the environmentally sustainable development as they are the leaders. He further points out the change of the sustainability concept. Earlier it meant to be sustainable in every aspect but has changed to be about environmental questions. As it is of great significance of today's society he welcomes it but means it is important to consider all aspects.

4.2.4 Summary

Two of the three companies interviewed has implemented a strategy for dealing with and contribute to sustainable development. Shell and Statoil are big businesses listed on the stock-market, whereas Preem is a smaller company situated only in Sweden. All three companies are exporting their products. Shell and Preem has quite the same idea about the definition of sustainable development. They mention the three pillars economy, society and environment. Though, Shell is focusing mostly on the society with their "future generation" compromise, whereas Statoil is trying to focus as balanced as possible between the three concepts. Preem's ambition is to contribute to future wants and needs and the development of the oil business.

The feedback from the stakeholders has been both good and bad. Statoil gets mostly positive feedback whereas Preem rarely gets any, except from researchers, universities and politicians. Lindh at Shell points out the co-operation between the society and businesses which is an ongoing process and the importance for government intervention. All three companies agree the stakeholders are affecting what decisions to make.

The companies further agree on the economic benefits of sustainable development. Both Shell and Preem has made investments in order to become more environmentally friendly. Shell, Statoil and Preem also sees competitive advantages with the concept of sustainability. Statoil think they are ahead in the big questions whereas Preem think they are the leaders with less harmful products and Shell sees advantages with customer satisfaction.

The focus of the three companies lies mainly at climate change, but Shell is focusing firstly on the handle of waste and Preem on carbon dioxide. Both Shell and Statoil is working in global networks and comments the concept of sustainability to be a big issue world wide. As Preem is owned by a private person and not listed on the stock-market the development of the strategy has been delayed compared to other companies, but the concept is very actual these days as the company is trying to implement a complete strategy into their work.

As found of the interviews there is a positive feedback of the sustainability concept and all three companies see the connection between environmental sustainability and competitive advantage. However, Egset at Statoil points out the significance of sustainable development as it brings long term interest and make companies recognise their responsibility and claims the concept should not be used as competition means but as a framework for a better future living.

5. Analysis

In this section, the authors will analyse the collected empirical data. The empirical data was categorised in the last section according to the respondents' point of view but in this section it will be analysed in different subcategories. The interviewees' point of view and response is viewed in the previous section.

5.1 Sustainable Development

The empirical findings of possible sustainable development are subcategorised into economical view, environmental view and social view in order to compare it to the theory in a more relevant way.

5.1.1 Economical view

When the Economical view was examined, the authors could find many similarities between the theoretical data and the empirical data.

According to Lindh from Shell, the company cannot gain any economic advantage in the long term if they do not implement the sustainable development strategy and by implementing this strategy, the companies can gain competitive advantage. In addition, it is very important to maintain costumers' satisfaction. In parallel, it is viewed in the theory that the sustainable development and the economic view are the same when it comes to limited resources such as oil; gas etc. as the adoption of sustainable development strategy has become more important for companies not only to be environmental friendly but to gain competitive advantage as well. Lindh also agrees that the sustainability strategy adds value to the company and pressure the company to become more responsible towards people and their needs.

Göran Byström at Preem states that they do not have much information about sustainable development strategy; nevertheless, the company has adapted another way for applying this strategy indirectly. They gain competitive advantage by satisfying the customers through providing them with less harmful products and developing the products continually. Though they have not applied the sustainable development strategy, they do their best to protect the environment in other ways.

As a result, the authors have found that every company has applied this strategy in a different way, some of which can be applied to the theory and the others do not, but at the end all these different strategies give nearly the same result as it is viewed in the theory. The theory indicates that the environmental dimensions can be reduced not only by implementing the sustainability strategy, but also in other manners such as economic growth which leads to environmental benefits. As one of PESTEL's external factors implies (the economic influence on organisations) the companies have to be more environmental friendly and adapt to the new strategies that reduce the disposal impact on the environment and thereby could gain benefit from new economic and social processes (Elliot, 2006). The companies have also to take into consideration how to handle the disposal and waste, which affect the environment (see section 1.1) and; therefore; they have to investigate which positive outcomes the sustainable development would bring to both the company and the environment.

Moreover, the respondents also claims that the companies are not paying any taxes on disposal but they can be fined if they handle the disposal incorrectly. Therefore, they are following this manner for protecting the environment, society and providing customers' satisfaction. According to the theory, an economic barrier which could appear is the inability of companies to recognise concealed costs of waste, such as treatment and disposal charges, insurance and potential future environmental legal responsibility (Barnes and Barnes, 1999). As noticed the companies are aware of the hidden costs and are making corrections as for example investments in more environmentally friendly products. The investments are expensive but it brings long term benefits.

The respondents indicate that the companies are trying to develop new products which are more environmentaly friendly and which could increase their earning further in the future. They aime to use these products as a meanse to increase the competitve profitability.

Furthermore, one of their financial objectives is to ensure long term value creation. The great challenge in long term value is to ensure the balance between the profitability and production growth. Additionally, Preem is working closely with stakeholders as for example governments to develop the product life-cycle in line with governmental policies. The companies are focusing on developing new business opportunities which create added value for the companies. With this develpment, the primary aim of companies is to create value sthrough profitability and sustainable business without causing harm to the people or to the environment. According to the theory presented in this study, it is noted that or achieving acceptable profit margin, the cost cutting is as important as the production volume. Moreover, the importance of producing is not only found in the way the products are being produced or how the waste is handled, but to take into consideration the whole life-cycle of the production process. The life-cycle analysis should describe and look into relevant environmental effects such as the handling of raw materials, production, trade, use, recycling, waste handling and transportation between each stage. A link has been acknowledged between economic growth and the change of demand. When people gain more, they demand better products (Industriförbundet, 1998). Many businesses see the environmental challenge as a threat and fail to realise the opportunities required for developing new markets, new products and new processes. In the short term, there is a need to look at the business environmental impacts and to find ways in which these can be made acceptable (business responsibility). The sustainable development strategy helps companies to develop opportunities and manage economic, environmental and social risks. Many investors consider it as a definitive value for success (Cheney, 2004, p. 14; Hart and Milstein, 2003, p. 57).

As a conclusion, the companies further agree on the economic benefits of sustainable development. Both Shell and Preem has made investments in order to become more environmentally friendly. Shell, Statoil and Preem also sees competitive advantages with the concept of sustainability. Statoil think they are ahead in the big questions whereas Preem think they are the leaders with less harmful products and Shell sees advantages with customer satisfaction. The linkage between theory and data is clear and there are advantages to gain by the economic view.

5.1.2 Environmental view

The environmental view will be studied by the authors in order to compare the findings with the theory data.

According to the theory, during the 1970s several member states of the European Union started to have control on the increasing of pollution problems from different natural resources with national policies (Barnes and Barnes 1999). Furthermore, the governments and the industries became more aware of the impact that disposals had on air and water. Therefore, the governments and the industries realized that by implementing a strategy, which could control the waste and the pollution, would gain environmental sustainability and provide a clean nature for the future generations (DiPeso, 2000). In parallel, the respondent in Shell states that the company is trying to control the waste and to remove it completely, therefore the company has built secondary containment for the oil so that the products are safer and to reduce the spill. Lindh says that this costs a lot but it brings a competitive advantage to the company at the same time. When it comes to Statoil, Fjæran points out that the environmental sustainability policy has a strong impact on the company’s activity. The most important issues for the company are climate change mitigations, water resource management, oil spill response, discharges to sea and waste management. According to Sjöberg the main environmental concern is the carbon dioxide which have a negative impact on the environment and this has to be taken into consideration in order to have a clean environment in the future. Therefore, the company is working to minimize these impacts.

Furthermore, the external factors from PESTEL emphasise the environmental protection. As noticed, people have become more aware of certain products that cause damage to the environment. These damages are caused during the process of manufacturing, use or disposal of products (Rock, 1989).

Mawhinney states that the awareness of environmental protection increased among people globally during 1960-2000 which caused an environmental change. The people has always had a desire to control the environment and the most important step at this time is to handle the effect of climate change. Furtheremore, it is important to handle disposals and toxic materials (2002). Lindh claims that Shell’s focus lies on climate change because it is important for the company to be environmental friendly. As Fjæran states that at the moment the climate issues are the dominant aspect globally, even though all the aspects are important. In addition, Sjöberg says that the climate is the most important query at the moment and in the future he thinks there is a need for a continuous effort on the longer run to handle the impacts.

As a result, the focus of the three companies lies mainly at climate change, but Shell is focusing firstly on the handle of waste and Preem on carbon dioxide. Both Shell and Statoil is working in global networks and comments the concept of sustainability to be a big issue world wide. As Preem is owned by a private person and not listed on the stock-market the development of the strategy has been delayed compared to other companies, but the concept is very actual these days as the company is trying to implement a complete strategy into their work. Once again the linkage between theory and empirical data is strong and the importance of the environmental view is verified.

5.1.3 Social view

The social view has had its impact also on the company's decision-making and the environment protection. Therefore, the authors will analyse the empirical data.

According to the theory, the sustainable development strategy is a well known social guiding model which integrates the economic, social and environmental issues in all levels in the short and long terms. Actually, this strategy ought to be followed by everybody in a different way. The governments contribute to the sustainable development policy and it is often involving some sorts of regulatory force, which the managements of companies should more or less willingly apply. With the term "more or less willingly", stakeholders influence come into play and they can affect the decision making in any company. Stakeholders (including governments) are important for transmitting the sustainable development strategy from society groups to the world business (Steurer et al, 2005). Furthermore, according to the PESTEL analysis the society and stakeholders have its impact on the organisations. As Elliot states, more government pressure on the companies by enforcements or penalties is anticipated, if they do not apply the imposed regulations. Therefore, the companies have to be environmental sustainable in order to avoid facing penalties (2006).

As of the empirical data, Lindh at Shell states that the society on the one hand is the ones with interest in the company and their influence means the company can go on and on the other hand, a clean environment will provide a better living for people, which is an important aspect of the sustainable development strategy of the company. He claims also that the society has the possibility to impact on the company. It is important for the company to listen to different stakeholder views of how to handle each step. Fjæran at Statoil claims that there was no specific pressure related to the implementation of the sustainable development strategy but he feels there will always be a pressure from stakeholders. Further on, Fjæran believes the company has to stay environmentally friendly wherever they operate. It is a must for the oil and gas industry and he thinks it brings competitive advantage to the company to be ahead in the aspects of the environment. In addition, he says that the variety of stakeholders affects the decision-making in the company. Sjöberg at Preem states that the stakeholders are very important in the decision-making of the company. Further on, the society and politicians are affecting the decision that the company take.

Through using an stakeholder analysis, there are seven actors, which are going to have an affect on an organisation. These actors are: costumers, suppliers, government, managers, employees, shareholders and banking institutions (Lynch, 2006). Each of these actors has to be taken into consideration while implementing new strategies, since either good or bad decisions are going to affect the organisation in different ways. The larger the company, the more the priorities of the stakeholders and the organisational goals may not correspond (Worthington and Britton, 2006). The significance of taking the views of the stakeholders into consideration can hereby be confirmed. Many decisions should be considered after a consultation with the stakeholders, but it must be mentioned that not all decisions can or will be. Though, the pressure from stakeholders will guide the companies to make the right decisions.

5.1.4 Sustainable development

From what can be found in the empirical data, Lindh at Shell states that the sustainable development strategy is adding value to the company as it is a stable base on which the company rely on to become more responsible towards people and their needs. When it comes to Statoil, Fjæran says that the concept is about contributing to a sustainable development through creating positive economic result; develop the society and not harming the environment. This is the way the company deals with sustainable development in all its activities. Furthermore, Martin Sjöberg, explains that Preem has not yet implemented a sustainable development strategy but has been introduced to it and is working towards one. The company’s ambition is to be able to contribute to future wants and needs and to be part of the future development of the oil business.

As in the theory Giddings *et al.* states, the concept of sustainable development is a wide range of meaning which is seen from different points of view by business men, governments, environmentalists and social reformers (2002). Furthermore, through protecting the environment and preventing pollution, companies would be able to make more profit, minimize risks, cutting disposal cost, gain competitive advantage and at the same time provide costumers higher quality products (DiPeso, 2000).

Egset at Statoil explains that by being sustainable to the environment the company can gain competitive advantage. As Lindh at Shell believes, the sustainable development strategy is helping the company to gain a competitive advantage and it is therefore important to stay sustainable. According to Sjöberg at Preem, a sustainable development strategy would add value to the company and it would increase the profit and the competitive advantage since it would allow some companies to become market leader. He adds, although the company does not have an entirely designed sustainability concept, it still can gain competitive advantage but it would gain more with the complete strategy. In the theory, Papmehl believes that the adoption of a sustainable development strategy has become more important for companies as they cannot only be environmentally friendly but also gain a competitive advantage. Though to be competitive, the companies need to find the edge in how to control the pollution, and to learn continuously, transform and be innovative in order to reduce the environmental impact, create social value, and to go beyond its competitors (2005).

The responder claim that through providing a cleaner, safer, and more sustainable world and by supplying chain relations which match the business, customers will gain competitive advantage. Therefore, companies have been trying to find ways of gaining competitive advantage through product development. The theoritical part of this study sugests that a business market grows and flourishes when the customers demand for more environmentally friendly products and the companies take such demands seriously and devote a considerable amount of money for research to meet such demands. In fact, the study has found that the companies covered by this study are doing their best to use the most developed technologies to create new products which are environmentlly friendly and closely meet the demands of their customres. This will further result in a business opportunity, when business realises the competitive advantage which could be reaped (Industriförbundet, 1998).

As it is viewed in the PESTEL analysis about competitiveness, companies should consider and understand the environmental changes, which have occurred lately and seize it as a key factor to create a competitive advantage by implementing a new sustainable strategy (Walsh, 2005).

The linkage between theory and empirical data has been shown through all the analysis and the concepts economic, environmental and social views has been verified to be significant in the work for a sustainable development strategy. The competitive advantage could be gained as well.

6. Conclusion

In this section there will be room for an outlined conclusion and comments to this. The authors will discuss the analysing and the outcomes of the study in order to find out whether the thesis has been answered and/or if the purpose has been achieved. Last but not least, concrete suggestions are given on how to further find incentives to study the area under discussion in a deeper manner.

6.1 Comments

The basic interest of this study arises from an awareness and curiosity of the growing problem of environmental harmfulness. The authors delved deeper into the subject and found that Sustainable Development was a frequently used word and; therefore, wanted to know whether it was applicable and useful for companies. The choice of oil companies came from the idea that those were the ones most hazardous to the environment as it is clear that the treatment and use of oil is having a negative impact on the environment. This was given both positive and negative reactions from the interviewed objects as their opinion of oil was that oil is the most important source of energy (fuel) for many individuals and companies but they realised its negative impacts on environment as well. The answer to this was to come up with less harmful products through differentiation and development of existing products.

As the study emerged the authors gradually gained more knowledge about the subject and a realisation of the significance of an environmentally sustainable strategy was learned. The response from the case studies was positive and; therefore, the authors found that a genuine contact was made. As the collected data was compared to the already excisting data, the authors found many similarities and the knowledge that a sustainabile strategy is inevitable, which is a prove of a high credibility. The authors find the implementation of a sustainable development strategy is of great importance for all companies in order to be environmentally friendly. As it will further bring competitive advantage, it can be seen as a trigger for companies, which have not implemented a strategy to implement it in the future.

The authors believe that an implementation of an environmentally sustainable strategy should be supported by the government but the incentive must come from the companies themselves. The government need to set boundaries for the companies to work against. Frameworks make it easier for each company (and the society) to know that the goals are reached. Furthermore, it is important with policies that reward the companies in order for each company to be treated equally in terms of actions.

6.2 Conclusion

As oil companies have in fact a greately negative impact on the environment and are constantly in need of developing their products, they can be more sensible to the study than other companies which are not in the same situation. The differentiation of more or less hazardous companies can have an impact on the outcome of this study. However, the authors think it is important to consider the hazardous companies as they are the ones of greatest importance of the significant questions raised in this study. Furthermore, it was not the incentive to generalise the study but to find a linkage between the primary and secondary data. The authors find that the collected primary data more or less match the theory data and the two chosen strategies (PESTEL and stakeholder analysis).

From the collected data, the authors could find that by implementing the sustainable development strategy, the companies can gain competitive advantage and be environmentally friendly at the same time and gain their stakeholders satisfaction. As it is found from the interviewed companies, the competitive advantage could be gained in other ways but it would increase the competitiveness by using a complete sustainability strategy.

The interviewed companies are trying to improve their products and processes continually in order to contribute to a safer environment and satisfy the customers' needs. The development leads to competitiveness as the companies are trying to be the first between the competitors which furthermore reap advantages.

The companies agree that a sustainable development strategy is an inevitable process in the future which all companies must implement. Moreover, the companies have benefited from the sustainability strategy which can verify the correspondence of the study.

6.3 Implications

The purpose of our study was to examine how the oil companies can gain competitive advantage via implementing a sustainable development strategy. The outcomesof the study according to the authors have identified these incentives and hopefully made it possible for such companies to notice possibilities with sustainability.

As the authors have stated in the conclusion of this study, the competitiveness can be gained in different ways, for example Preem could gain competitive advantage and be environmentally friendly via product differentiation. We believe that if companies apply this strategy and try to be more environmentally friendly, they can gain more benefits such as economic and social and provide a safer environment for future generations.

6.4 Discussion and critique

The authors have focused mainly on the positiveness of implementing a sustainable strategy. Therefore, the negative side might have been put aside and the result of the study may have been presented in a one-side aspect. However, only one major negative impact has been found which is the cost for improving the safety of the production performance. The cost is not seen as a burden by the companies but as a long-term investment for the future.

The way of gaining competitive advantage which is presented in the interviewed companies, has been viewed in the conclusion above. The conclusion clearly shows that the purpose of our study has been fulfilled. The positive outcomes in relation to the main purposes also put forward the possible incentives for other companies to become sustainable in their businesses. We wish to emphasise that the success of the companies which has implemented a sustainability strategy does not solemnly depend on their sustainability policy. Other companies may fail due to lack of feasibility with their overall strategy. However, this goes beyond the scope of this study.

6.5 Proposal for further research

It could be useful to put more resources into the research and even more objects/cases as they could affect the validity and reliability as to the same result or it could reject the thesis. The choice of companies from only one sector could have had an impact on the end result in which it would be valuable to extend it to a wider range of sectors.

There are many ways in which the environment can be affected and it could be inspiring to examine whether different aspects need more or less sustainable strategies.

Further a quantitative approach might have given the end result a different outcome and could thus be useful to be taken into consideration.

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Interviews

Byström, G., Chief Accountant, Preem Sweden, 18th April 2007, 9.00hrs CET.
Sjöberg, M., Environmental co-ordinator, Preem Sweden, 20th April 2007, 10.00hrs CET.
Lindh, P. O., Environmentally responsible, Shell Sweden, 7th May 2007, 9.00hrs CET.
Fjaeran, T., Senior vice president Environment, Statoil Norway, 7th May 2007, 12.00hrs CET
Egset, W., CSR Manager, Statoil Norway, 7th May 2007, 13.30hrs CET

Appendices

Appendix 1 – Interview Guide (Brief)

The authors have a great interest in the influence of implementing the Sustainable Development strategy on the entire organisation and its effect on the organisation's economy. Anyhow, we have chosen to focus on some particular area which we are especially interested in.
The issues which we are interested in to examine are:

- Background and the definition of the sustainable development concept.
- Competitiveness and the sustainable development strategy.
- Stakeholder relationship. Feedbacks which come from different stakeholders such as government, society, customers and the employees.
- The environmental view.
- The advantage and disadvantage of not implementing this strategy.
- Evaluation tools.

Furthermore, we are of course interested in any more information which concerns environmental and social view about implementing sustainable development strategy in any organisation. Finally, we wish to emphasise that we are seeking to find out the affect of the sustainable development strategy on above issues.

Appendix 2 – Interview Guide

Personal information:

Name:

What position do you have in the organization?

How long have you been in the company/this position?

Overall:

Has the company adapted the Sustainable Development Strategy?

Which are the main factors that underlie the Sustainability policy?

What does Sustainable Development mean to you and the company? Define?
This is due to the different literature definitions of SD.

How do the company involve in the development of the different sustainability processes?

In which way has the company changed since the introduction of SD?

Stakeholder concerns:

Are different stakeholders aware of the company's sustainability policy? If so, what kind of feedback have you received from them concerning your sustainability program?

Can the stakeholders affect the decision-making of the company? If so, how?

Economic view:

Will it be an economic issue for a company if it does not implement SD strategy? (Based on your experiences)

If so, how can SD influence the budget?

What is the level of tax on disposals?

Has the company made any market developments?

Has the company made any product developments? If so, What products did the company develop?

Are the products environmentally friendly? In what way?

Has the company made any changes of distributors?

How can/has these affected the company?

How did the company carry these out?

Has the company faced any fines either before or after implementing SD?

Has the company made economic growth by adapting to SD?

What laws/regulations have been introduced lately?

Have these laws/regulations affected the company's economy?

Have there been any costs to the company because of this?

Environmental view:

Which impact does the environmental sustainability policy have on this company?

Which are the main environmental concerns that the company focus on? Why these? Do you think these will change in the future?

What pressures did the company face before the SD strategy was implemented?

How important is it for the company to maintain environmentally friendly?

Social view:

How much impact can the society have on this company?

Sustainability:

What is the driving force behind sustainable development in the company?

What problems did the company face before implementing the SD strategy?

Are there any issues today?

Which concrete actions does the company take to achieve the sustainability policy?

Does the SD strategy add value to the company? If yes, how?

How does the company work with the SD strategy?

How sustainable is your company compared to other oil companies?

How far can this oil company push its sustainability policy?

Competitive Advantage:

How do you define the company's view of the concept Competitive Advantage?

Do you feel that the company competes with other oil companies about being sustainable to the environment? If yes, why?

Can the company gain competitive advantage by implementing this strategy (SD)?
How important is it for the company to stay sustainable?

How important are costumers' satisfaction for this company?

In which way does Competitive Advantage affect the development of the company's result?

Do you want to add anything else, which might be useful for this study?